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X-ray shielding device

Abstract

An X-ray device includes: an X-ray source configured to emit X-rays; a scintillator, the scintillator being configured to emit light in response to absorption of the X-rays; a detector; a frame enclosing the X-ray source and the detector; standoffs positioned on the frame; shielding panels comprising lead; one or more brackets with fasteners configured to attach to the standoffs; and exterior panels with fasteners configured to attach to the one or more brackets. The standoffs form a datum structure for the one or more brackets. Each of the standoffs has a length and a cross-sectional size, and each of the shielding panels includes holes having a cross-sectional size greater than the cross-sectional size of the standoffs by an amount that ensures the standoffs will pass through the holes. The length of the standoffs is greater than a maximum thickness possible for the given shielding panel due to variations in thickness.

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Background/Summary

BACKGROUND

(1) This specification relates to X-ray computed tomography (CT) devices.

(2) X-ray computed tomography (CT) is a technique that can be used by manufacturers in order to determine the quality of the products which they produce. X-ray CT is particularly useful to give manufacturers the ability to inspect certain parts of their products in a non-invasive, non-destructive fashion. Given this, X-ray CT is becoming more popular in production manufacturing settings where quality control is of high importance.

(3) X-ray CT devices require shielding to prevent harmful X-rays from leaking from the X-ray CT device.

SUMMARY

(4) This specification describes technologies relating to an X-ray shielding device. In particular, this specification describes systems and apparatuses configured to allow shielding of X-rays to prevent the emission of harmful X-rays.

(5) Compared to other metals and materials in general, lead is one of the most effective X-ray shielding materials, owing to its large attenuation coefficient and high density. Additionally, lead is relatively inexpensive compared to other X-ray shielding materials. Lead, however, is toxic to humans and other species, and poses other engineering problems. For example, due to its softness, obtaining a precisely aligned structure composed of lead can require iterative assembly processes, which can be slow and cumbersome. Additionally, since lead is heavy, panels of lead making up the exterior of an X-ray shielding device might require being split up into smaller panels for assembly. The panels being split up into neighboring, smaller panels create gaps in the X-ray shielding device, which increases the chance of X-ray radiation leaking from the X-ray shielding device.

(6) The described X-ray shielding device overcomes the challenges of working with lead through use of a datum structure to provide accurate mounting features that pass through the lead, thereby avoiding the use of lead for providing structural support. Additionally, in some implementations, a C-channel (or similarly shaped) structure within the X-ray shielding device can seal gaps between neighboring lead panels and reinforce an interior frame structure of the X-ray shielding device.

(7) In general, innovative aspects of the subject matter described in this specification can be embodied an X-ray device that includes: an X-ray source configured to emit X-rays; a scintillator arranged to absorb the X-rays after interaction with an object that has been placed in the X-ray device, the scintillator being configured to emit light in response to absorption of the X-rays; a detector arranged to receive the light from the scintillator; a frame enclosing the X-ray source and the detector; standoffs positioned on the frame; shielding panels comprising lead; one or more brackets with fasteners configured to attach to the standoffs; and exterior panels with fasteners configured to attach to the one or more brackets. The standoffs form a datum structure for the one or more brackets. Each of the standoffs can have a length and a cross-sectional size, and each of the shielding panels can include holes having a cross-sectional size that is greater than the cross-sectional size of the standoffs by an amount that ensures the standoffs will pass through the holes despite variations in a placement of the holes in a given shielding panel resulting from manufacturing of the given shielding panel. The length of the standoffs can be greater than a maximum thickness possible for the given shielding panel due to variations in thickness resulting from the manufacturing of the given shielding panel.

(8) Another general aspect can be embodied in an X-ray device that includes: an X-ray source configured to emit X-rays; a scintillator arranged to absorb the X-rays after interaction with an

object that has been placed in the X-ray device, the scintillator being configured to emit light in response to absorption of the X-rays; a detector arranged to receive the light from the scintillator; a frame enclosing the X-ray source and the detector; shielding panels including lead; a metal piece having fasteners configured to attach the metal piece with the frame; and the shielding piece being sized and positioned to prevent X-rays from passing (i) through the gap and (ii) through the holes in the first and second shielding panels on either side of the gap, which are usable when coupling the first and second shielding panels with the frame. Each of the shielding panels can include holes usable to couple the shielding panel with the frame, where a first and a second of the shielding panels protect a single side of the frame and have been reduced in size to facilitate installation of the first and second shielding panels. A gap can remain between the first shielding panel and the second shielding panel when installed on the single side of the frame, and the metal piece can be shaped to receive a shielding piece including lead.

(9) These and other implementations can each optionally include one or more of the following features. In some implementations, the X-ray device includes at least one additional shielding panel without lead.

(10) In some implementations, the X-ray device further includes at least one corner guard including an angled sheet of lead extending between two adjacent sides of the frame. The angled sheet of lead can be placed at an acute angle between the two adjacent sides of the shielding panels, and edges of the angled sheet of lead can be chamfered in accordance with the acute angle.

(11) In some implementations, the X-ray device further includes a metal sheet on which the corner guard is attached. The metal sheet can be attached to the frame

(12) In some implementations, the shielding panels comprise at least one laminate comprising lead and steel. In some implementations, each of the shielding panels is a laminate comprising lead and steel.

(13) In some implementations, at least one of the one or more brackets is configured and arranged to have two of the exterior panels located on different sides of the X-ray device attached to a same bracket.

(14) In some implementations, the X-ray device includes one or more additional shielding panels including a plastic impregnated with lead-free particles.

(15) In some implementations, each of the standoffs includes a threaded hole, and each of the fasteners is configured to attach the brackets to the standoffs is a bolt configured to mate with the threaded hole.

(16) In some implementations, each shielding panel of the shielding panels weighs less than 39 kilograms.

(17) In some implementations, the cross-sectional size of the holes is greater than the cross-sectional size of the standoffs by about 35%. In some implementations, a cross-sectional size of the threaded holes is greater than a cross-sectional size of the standoffs by about 35%.

(18) In some implementations, a shape of the holes in the shielding panels is a rounded rectangle.

(19) In some implementations, the standoffs are arranged on at least two sides of the frame, thereby forming the datum structure in at least two intersecting planes of three-dimensional space.

(20) In some implementations, neighboring edges of first and second shielding panels of the shielding panels form a gap. The X-ray device can further include: a metal piece having fasteners configured to attach the metal piece with the frame; and the shielding piece being sized and positioned to prevent X-rays from passing (i) through the gap and (ii) through holes in the first and second shielding panels on either side of the gap, which are usable to couple the first and second shielding panels with the frame. The metal piece can be shaped to receive a shielding piece including lead.

(21) In some implementations, the X-ray device further includes additional shielding disposed on the shielding.

(22) In some implementations, the X-ray device further includes corner guards including angled

sheets of lead extending between two adjacent faces of the shielding panels. The corner guards can be angled at an acute angle between the two adjacent faces of the shielding panels, and edges of the corner guards are chamfered at the acute angle.
(23) In some implementations, the X-ray device further includes a sheet on which each corner guard of the corner guards is attached. The sheet can be attached to the frame.
(24) In some implementations, the X-ray device further includes brackets and standoffs. The standoffs can include threaded holes, and fasteners that attach the brackets to the standoffs can be bolts configured to mate with the threaded holes in the standoffs.
(25) Particular embodiments of the subject matter described in this specification can be implemented to realize one or more of the following advantages. The systems and apparatuses described can provide highly efficient X-ray shielding, while preventing direct contact with lead by a user. Further, the design and assembly process can be quicker and less expensive by avoiding iterative processes for assembling the X-ray shielding device.
(26) In some implementations, the systems and apparatuses described can be assembled without iterative processes, thereby reducing resources, e.g., equipment and time, to create a CT device and increasing a scalability of the CT device. For example, the exterior panels can be more robustly mounted with reduced clearance compared to assemblies without the datum structure. The more robustly mounted exterior panels can have a reduced likelihood of shifting during shipment or use. As another example, the CT device can accommodate larger objects for scanning if the size can be increased without increased error in alignment.
(27) The details of one or more embodiments of the subject matter described in this specification are set forth in the accompanying drawings and the description below. Other features, aspects, and advantages of the invention will become apparent from the description, the drawings, and the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1**A shows a perspective view of an example of an X-ray shielding device.
(2) FIG. **1**B shows a cross-sectional and schematic view of an example of an X-ray computed tomography system within the X-ray shielding device of FIG. **1**A.
(3) FIG. **2**A shows a perspective view of an example of a frame and a datum structure of the X-ray shielding device of FIG. **1**A.
(4) FIG. **2**B shows a perspective view of an example of exterior panels of the X-ray shielding device of FIG. **1**A with a door.
(5) FIGS. **3**A-**3**C respectively show perspective, cross-sectional, and close-up views of standoffs from the X-ray shielding device of FIG. **1**A.
(6) FIG. **3**D shows a cross-sectional view of a shielding panel of the X-ray device of FIG. **1**A held in place by fasteners.
(7) FIGS. **4**A and **4**B are perspective and close-up views of the brackets from the X-ray shielding device of FIG. **1**A.
(8) FIGS. **4**C and **4**D are cross-sectional views of the shielding panels and brackets of FIGS. **4**A and **4**B.
(9) FIGS. **5**A and **5**B show cross-sectional views of examples of a metal piece having fasteners configured to attach the metal piece with the frame within the X-ray shielding device of FIG. **1**A.
(10) FIGS. **6**A and **6**B show exploded views of the C-channel and additional shielding of FIG. **5**A.
(11) FIG. **6**C depicts a perspective view of the C-channel of FIG. **5**A riveted to a panel of the frame of FIG. **2**A.
(12) FIG. **7** shows a cross-sectional view of an example of a corner guard within the X-ray

shielding device of FIG. **1**A.
(13) Like reference numbers and designations in the various drawings indicate like elements.
DETAILED DESCRIPTION
(14) FIG. **1**A shows a perspective view of an example of an X-ray shielding device **100**. The X-ray shielding device **100** can enclose a system that generates X-rays and prevents the generated X-rays from escaping from the X-ray shielding device **100**. Panels made from lead, which is one of the most efficient X-ray absorbing materials, can surround the system. However, there are gaps **114** where panels are joined together, e.g., between panels on the same face **116** of the X-ray shielding device **100** and at corners **118** of the X-ray shielding device **100**.
(15) Although obtaining efficient X-ray shielding does not require the X-ray shielding device **100** to be perfectly sealed, e.g., since the X-rays may only travel in certain directions based on the geometry of the X-ray shielding device **100** and the enclosed system, adding shielding in front of the gaps **114** between panels of lead can improve the shielding efficiency. However, using lead to support structural components presents problems, due to lead's heaviness and softness. Accordingly, the present disclosure describes a datum structure for assembling an X-ray shielding device **100** that does not use lead for structural support, as well as a metal piece with additional shielding that also provides additional structural support.
(16) FIG. **1**B shows a cross-sectional and schematic view of an example of an X-ray computed tomography system **110** within the X-ray shielding device **100** of FIG. **1**A. The X-ray computed tomography system **110** includes X-ray source **101** configured to emit X-rays **102** towards scintillator **103**. As X-rays **102** pass through scan target **107** and collide with scintillator **103**, scintillator **103** absorbs the X-rays **102** and emits visible light **108**, which generates an image. In some implementations, X-ray computed tomography system **110** can include one or more mirrors, e.g., folded optics, to reduce the optical path length and thus size of the X-ray shielding device **100**.
(17) In various example implementations, the X-ray computed tomography system **110** may include motion system **106** configured to move, reposition, manoeuvre, or otherwise manipulate the detector **105** and/or scan target **107** relative to the X-ray source **101**. Furthermore, X-ray computed tomography (CT) system **110** may be an X-ray CT device. In various examples, X-ray source **101** may emit an X-ray beam, which may be a pencil beam, fan beam, cone beam, etc.
(18) The shielding efficiency can be defined as the total energy of X-rays that escape the X-ray shielding device divided by the total energy of radiation produced by the X-ray source. For example, in some implementations, X-ray shielding device **100** only lets escape about one millionth of the energy of the generated X-rays. In some implementations, the shielding efficiency is selected to comply with regulations of the United States Food and Drug Administration (FDA) with a margin, e.g., stricter than the regulations.
(19) FIG. **2**A shows a perspective view of an example of a frame **120** and a datum structure **130** of the X-ray shielding device **100** of FIG. **1**A. The frame **120** can include multiple sheets of metal, such as steel, that form six faces **121***a*, **121***b*, **121***c*, **121***d*, **121***e*, and **121***f* of the frame **120**. In some implementations, the frame has a rectangular prism shape, with top, bottom, and side faces.
(20) In some implementations, at least some of the faces **121***a-f* have indents formed by the sheets of metal not being completely planar, e.g., face **121***b* includes two sheets **125***a* and **125***b*, each of which include at least two 90° bends on the surface of the sheet. These bends can increase a stiffness of the sheets of metal, thereby turning them into structural components. At least some of the faces **121***a-f* can have an indent that creates space for fasteners between the frame **120** and exterior components.
(21) The sheets of metal forming faces **121***a-f* can each have holes **126**, through which fasteners can extend. In some implementations, the holes **126** are threaded to mate with threaded fasteners.
(22) The datum structure **130** is marked by the dashed lines to indicate the linear alignment of standoffs **128**, which in combination with the spacing between standoffs, makes the standoffs a datum structure **130**. In some implementations, the standoffs **128** can be pressed into sheet metal or

be blind, through hole, and/or threaded standoffs. For example, standoffs **128** can be pressed into sheet metal of the frame **120** by a sheet metal fabricator. With sufficient force, the standoffs **128** bond to the sheet metal and permanently remain in place. In some implementations, the standoffs **128** are threaded to mate with the frame **120**.

(23) The datum structure **130** defines various points for aligning the frame **120** with the panels of the X-ray shielding device **100** and other components such as brackets and rails, e.g., holes in each of the sheets of metal forming the frame and panels being aligned and components of the datum structure **130** passing through the holes.

(24) The datum structure **130** aids in the alignment of the exterior panels of the X-ray shielding device **100**, which will be discussed in reference to FIGS. **4**A-**4**D. In other words, the datum structure **130** can be three-dimensional, and the standoffs **128** forming the datum structure **130** can be arranged in at least three intersecting planes. In some implementations, the datum structure **130** can be arranged in fewer than three intersecting planes, e.g., two parallel planes or two intersecting planes.

(25) At least one of the sheets **122** making up a side face can include an opening **124** for a door for letting objects pass within and out of the X-ray shielding device **100**.

(26) FIG. **2**B shows a perspective view **200** of an example of exterior panels **138***a*, **138***b*, and **138***c* of the X-ray shielding device **100** of FIG. **1**A with a door **129**. The exterior panels **138** are made from a material other than lead, so that it is safe for human contact. In some implementations, the exterior panels **138***a-c* include metal, plastic, and/or other materials

(27) FIGS. **3**A-**3**C respectively show perspective and cross-sectional views **300***a*, **300***b*, and **300***c* of standoffs **128** passing through shielding panels **132** (which can include lead in various forms) from the X-ray shielding device **100** of FIG. **1**A. In perspective view **300***a*, the standoffs **128** pass from the frame **120** and through the shielding panels **132**.

(28) In some implementations, the shielding panels **132** include lead or a lead-free shielding material, e.g., a plastic impregnated with X-ray absorbing particles. For example, most of the shielding panels **132** can be pure lead or lead sandwiched between two layers of another material, and the remaining shielding panels can be lead-free. For example, lead-free, secondary shielding panels can cover portions of shielding panels **132** that include lead but have small leaks, such as holes or seams.

(29) Due to the high density of lead, panels made from pure lead can be prone to bending and drooping along the direction of gravity while being assembled. To reduce the weight of the X-ray shielding device **100** and to avoid uneven shielding panels **132**, the size and thus weight of the shielding panels **132** can be limited. For example, some or all the shielding panels **132** can be less than a limiting weight of between 20 and 40 kilograms, e.g., less than 23 or 39 kilograms. In order to maintain the weight of each shielding panel **132** below a threshold, a single face **131** can include multiple shielding panels **132**. The specific weight limit can depend on the average strength of the workers in a given country, the workers' health and safety regulations in a given country, or both.

(30) Although steel and lead provide important characteristics, e.g., structural support and X-ray absorption, respectively, manufacturing laminates of steel and lead can be less precise compared to other metals. Accordingly, providing a nonzero tolerance for interconnecting lead and steel laminates can facilitate aligning the lead and steel laminates

(31) The standoffs **128** pass through holes in the shielding panels **132**, where the holes are sized to provide a tolerance for the standoffs. For example, as depicted in cross-sectional view **300***b*, holes **134** in the shielding panels **132** can be slots with a 11 mm width W.sub.1 along the Y axis, and the standoffs **128** can have a rounded-rectangular cross-section with a 7.12 mm width W.sub.2 along the Y axis, providing a radial clearance C of about 2 mm relative to the slot-shaped hole **134**. In some implementations, the radial clearance is determined by percentage of the standoff **128**, e.g., the holes **134** in the shielding panel **132** are sized to provide a percentile radial clearance of the width of the standoff **128**, e.g., 35% of the width of the standoff **128**. In some implementations,

there is also a radial clearance for the standoffs **128** relative to the holes in the shielding panels **132** along a direction perpendicular to the width, e.g., the Z axis.

(32) Using slots in the shielding panels **132**, e.g., a rounded rectangle with a longest edge along the direction of the gap (Y axis in this example), can provide more lateral tolerance compared to using circular holes. If the cross-sectional size of the holes **134** is greater than the cross-sectional size of the standoffs **128** by a certain amount, e.g., 2 mm, the standoffs **128** will pass through the holes **134** despite variations in placement of the holes **134** in a given shielding panel **132** resulting from manufacturing of the given shielding panel **132**.

(33) As another example, each of the standoffs **128** and the holes **134** in the shielding panels **132** can have a circular cross-section, where the radius of the cross-section of the holes in the shielding panels **132** is greater than the radius of cross-section of the standoffs **128**.

(34) To connect the shielding panels **132** to the frame **120**, the standoffs **128** must be long enough to at least pass through the shielding panels **132**. As shown in cross-sectional view **300***c*, the standoff **128** has a first length L.sub.1 along the X axis that is greater than a second length L.sub.2 of the shielding panel **132** along the X axis, such that the standoff **128** can pass through the shielding panel **132** to connect the exterior panel **138** to the frame **120**.

(35) As indicated by the different widths along the X direction of an exterior end **128***a* of a standoff and an interior end **128***b* of a standoff, the exterior end **128***a* of the standoff can be long enough to attach to an exterior panel **138** through a bracket, while the interior end **128***b* of the standoff is nearly flat after having been pressed into the sheet metal of the frame **120**. In some implementations, the interior end **128***b* of the standoff has a polygonal, e.g., hexagonal, cross section, e.g., along the X axis in FIG. **3**C, to prevent rotation of the standoff **128**.

(36) In some implementations, the shielding panel **132** is a laminate, e.g., made of one or more materials, such as at least two materials bonded together to form a laminate. For example, shielding panel **132** can be a laminate formed by a lead layer **136***a* sandwiched by layers **136***b* with adhesive **305** at the interface between the lead layer **136***a* and the layers **136***b*. Layers **136***b* can be made from a material stiffer than lead, e.g., steel, to provide extra structural support. When the shielding panel **132** is a laminate, the length L.sub.1 of the standoff **128** can be greater than the second length L.sub.2 of the shielding panel **132** including the length of the layers **136***b*. When the shielding panel **132** is a single layer of lead, the length L.sub.1 of the standoff **128** can be greater than a third length L.sub.3 of lead layers **136***a*.

(37) In general, the length L.sub.1 of the standoff **128** is greater than the shielding panel **132**, including when the shielding panel **132** includes one or more laminates **136**, to ensure that the brackets for the exterior panels **138** do not touch the shielding panels **132**. Accordingly, the location of the brackets for the exterior panels **138** and the exterior panels **138** are not affected by variations in the thickness and/or flatness of the shielding panels **132**.

(38) The laminates **136** can be sheets substantially parallel to the shielding panels **132**. The cross-sectional size of the laminates **136** can be approximately the same as the shielding panels **132**, e.g., smaller than the shielding panels **132** by about 0.5 millimeters. In some implementations, the laminate **136** is attached to the shielding panel **132** with an adhesive **305**.

(39) The length of the standoffs L.sub.1 is greater than a maximum thickness possible for the given shielding panel, e.g., length L.sub.2, due to variations in thickness resulting from the manufacturing of the given shielding panel. In some implementations, the length L.sub.2 of the shielding panel **132** along the X axis can be relatively even, e.g., the same within 0.5 millimeters. Due to lead's softness, lead is relatively easy to machine lead panels of various thicknesses.

(40) FIG. **3**D shows a cross-sectional view **300***d* of a shielding panel **132** of the X-ray device of FIG. **1**A held in place by fasteners **135***a*. Fasteners **135***a* are located around the perimeter of each shielding panel **132** as depicted in FIG. **3**A. In FIG. **3**A, the standoffs **128** are represented by smaller circles, and larger circles correspond to washers **135***b* with a raised center corresponding to the fasteners **135***a*.

(41) The holes **134** in shielding panel **132** can have a variety of purposes. For example, some of holes **134** are sized and arranged to allow standoffs **128** to couple to the frame **120** to brackets **140** for mounting the exterior panels **138**, and some of holes **134** are sized and arranged to allow fasteners **135***a* to couple the shielding panels **132** to the frame **120**.
(42) As depicted in FIG. **3**D, a fastener **135***a*, such as a bolt, extends completely through the shielding panel **132** to attach to the frame **120**. In this example, the shielding panel **132** is a laminate with a lead layer **136***a* sandwiched by layers **136***b*. The washers **135***b* are on an exterior side of the shielding panel **132** and surround the fastener **135***a*. Washers **135***b* distribute the load of fastener **135***a* on the shielding panel. Fasteners **135***c* are on an interior side of the of the shielding panel **132** and surround the fastener **135***a*. In some implementations, fasteners **135***c* are nuts pressed into the sheet metal of the frame **120**.
(43) Given the difficulty with precisely manufacturing lead shielding panels **132**, the holes **134** in the shielding panels and fasteners **135***a* are sized to provide a clearance around the fastener **135***a*. In FIG. **3**D, clearances C.sub.1 and C.sub.2 are the vertical clearances, e.g., along the Z axis, above and below, respectively, between the fastener **135***a* and hole **134** of the shielding panel **132**. In some implementations, there is a clearance between the fasteners **135***a* and holes **134** in either one or both of the horizontal directions.
(44) FIGS. **4**A and **4**B are perspective and close-up views **400***a* and **400***b* of brackets **140** from the X-ray shielding device **100** of FIG. **1**A. The datum structure **130** provides mounting locations for precisely mounting brackets **140**, as well as door motion and locking components. The brackets **140***a*-**140***g* can hold the exterior panels **138** (pictured in FIG. **2**B) in place, e.g., attached to the frame **120** via fasteners, such as standoffs or bolts. Each bracket **140** can attach to at least two exterior panels **138**. Bracket **140***c* can include holes that precisely align with the datum structure **130**, such that standoffs **128** can extend from the frame **120** through the shielding panels **132** to mount the brackets. Using the datum structure **130** can also increase the repeatability of methods of accurately aligning the various components of the X-ray shielding device **100**.
(45) In some implementations, as depicted in close-up perspective view **400***b*, some of the brackets, e.g., brackets **140***a*, **140***b*, **140***f* and **140***g*, include edges **143***a* and **143***b* that meet at a right angle to attach exterior panels **138** that meet at a right angle. One of the edges, e.g., edge **143***a*, can be ridged. In some implementations, some brackets, e.g., bracket **140***g* can hold two exterior panels, e.g., exterior panels **138***a* and **138***c*, with different orientations while only being attached to standoffs **128**.
(46) In some implementations, the X-ray shielding device **100** includes door guide rails **142**, which are configured to allow a sliding door, e.g., door **129**, to slide along a horizontal direction, e.g., the X axis in this example, to both cover and uncover the opening **124**. Additionally, the X-ray shielding device **100** can include a door interlock **144**, which locks the sliding door into place when covering the opening **124**. When the sliding door is latched into place by the door interlock **144**, the sliding door is in a position that reduces X-ray leakage compared to other positions.
(47) FIGS. **4**C and **4**D are cross-sectional and close-up views **400***c* and **400***d* of the shielding panels and brackets of FIGS. **4**A and **4**B. As can be seen in both views **400***c* and **400***d*, the bracket **140** is attached to standoffs **128**, which in turn are part of the frame **120**. A small gap **145**, e.g., 1.5-3 mm, allows for variation in the thickness and flatness tolerance deviations of the shielding panel **132** without affecting the position of the bracket **140**. The size of the gap **145** depends on material thicknesses of the shielding panels **132** and length(s) of the standoffs **128**.
(48) FIGS. **5**A and **5**B show cross-sectional views **500***a* and **500***b* of examples of a metal piece having fasteners configured to attach the metal piece with the frame **120** within the X-ray shielding device **100** of FIG. **1**A. The metal piece provides mounting locations for the shielding panels **132** in addition to the mounting locations of the datum structure **130** at corners of the X-ray shielding device **100**.
(49) In some implementations, the metal piece is a C-channel **150**, e.g., a continuous piece of metal

and is called a “C”-channel for having a shape like the letter “C.” For example, the C-channel **150** can be composed of two parallel segments **151***a* **151***c* (extending in a first direction) connected by a segment **151***b* (extending in a second direction) perpendicular to both segments **151***a* and **151***c* and connected at the edges of segments **151***a* and **151***c*.

(50) In general, the metal piece can have various shapes as long as the shape creates room to receive the additional shielding **148** and includes at least one surface by which the metal piece attaches to the frame **120**. For example, the shape and location of the metal piece can be selected to cover holes **134** in the shielding panels **132** for fasteners **135***a* and standoffs **128**.

(51) An advantage of using the C-shape is there are two locations where the metal piece attaches to the frame **120** on either side of the additional shielding **148** e.g., to the left and right of and below additional shielding **148** in FIG. **5**A. As a result, a C-shaped metal piece adds structural support and strength to the frame **120** and facilitates holding of the weight of the main shielding panels **132**.

(52) The C-channel **150** can have associated fasteners, e.g., fastener **154**, configured to attach the C-channel **150** to the frame **120**. In some implementations, the fastener **154** passes through two aligned holes of the C-channel **150** and the frame **120**.

(53) The C-channel **150** can be disposed between shielding panels **132** and the frame **120** where gaps **147** between neighboring shielding panels **132** exist. Due to the shape of the C-channel **150**, e.g., having two segments extending in the direction from the frame **120** to the shielding panels **132** (along the Y axis in this example) and another segment parallel to the direction of the gap **147** between the shielding panels **132** (along the X axis in this example), additional shielding **148** can be located between the frame **120** to the shielding panels **132** and block the gap **147**.

(54) The additional shielding **148** can be attached to a plate **152**, which is configured to attach to the C-channel **150**. For example, the plate **152** can define a hole through which a fastener **156** can pass and attach to the segment **151***b* of the C-channel **150**. The plate **152** can be made of a metal stiffer than lead, e.g., steel. In some implementations, the additional shielding **148** can be directly attached to the C-channel **150** without the plate **152**. However, when the additional shielding **148** includes lead, creating precise holes for fastening can be easier when using a stiffer metal compared to lead.

(55) FIGS. **6**A and **6**B show exploded views **600***a* and **600***b* of the C-channel **150** and additional shielding **148** of FIG. **5**A. FIGS. **6**A and **6**B depict the C-channel **150** and additional shielding **148** but flipped about the Y axis. As depicted in FIGS. **6**A and **6**B, the additional shielding fits between the two segments **151***a* and **151***c* of the C-channel **150**. Holes **153** in the additional shielding **148** are aligned with holes **155** in the C-channel **150**, such that fasteners **156** can connect the additional shielding **148** to the C-channel **150**.

(56) As visible in exploded views **600***a* and **600***b*, each of the first through third segments **151***a*-**151***c* extends in the third direction perpendicular to both the first and second directions, e.g., a vertical direction along the Z axis in this example. The C-channel **150** can be long enough in the vertical direction to cover an entire gap between neighboring shielding panels **132**.

(57) FIG. **5**B depicts a similar view as FIG. **5**A, with some differences. First, this view is for different side of the X-ray shielding device **100**, and therefore the orientation is flipped along the X axis. Second, the C-channel **158** extends longer along the X axis compared to C-channel **150**. In this example, C-channel **158** extends far enough along the X axis to span the opening **124** for the door **129**. With reference to FIG. **2**A, the C-channel **158** is a part of the sheet **122**, which includes an opening **124** for the door **129**. Third, the C-channel **158** is attached to an additional layer of shielding **149**, which contacts the exterior panels **138**.

(58) FIG. **6**C depicts a perspective view **600***c* of the C-channel **150** of FIG. **5**A riveted to a panel **127** of the frame **120** of FIG. **2**A. The C-channel **150** being riveted to the panel **127** can provide structural reinforcement for the frame **120**. In some implementations, the C-channel **150** is riveted to the panel **127** using flanges on both the top and bottom, e.g., along the Z axis, of the C-channel **150**.

(59) FIG. **7** shows a cross-sectional view **700** of an example of a corner guard **160** within the X-ray shielding device of FIG. **1**A. Similarly to how the additional shielding **148** prevents strays X-rays from escaping through gaps between neighboring, parallel shielding panels **132**, the corner guards **160** can absorb X-rays that would otherwise pass through gaps between neighboring, perpendicular shielding panels **132**. The corner guards **160** can include an angled sheet of lead **168** or another shielding material.
(60) In some implementations, the corner guard **160** includes a sheet **162**, which is made of a metal stiffer than lead, such as steel. The sheet **162** can include holes configured to accept fasteners **163**, which extend through both the sheets **162** and a member **164**. The member **164** can have an open rectangular shape, e.g., a rectangle with portions from two adjacent sides missing at the point where the two adjacent sides would have met. The metal frame can include portions **166** that extend from neighboring, perpendicular panels of the frame **120**, e.g., an L-shaped portion. The portion **166** can be shaped to receive a sheet **162**, which is attached to member **164**, of a corner guard **160**. For example, the angled edges **161***a* and **161***b* of the angled sheet of lead **168** can be parallel to each of the L-shaped portions **166**. The fasteners **163** can extend through the sheet **162**, the member **164**, and the portions **166** to attach the corner guards **160** to the frame **120**. The sheet **162** and member **164** can be fastened together to form a more rigid and structural composite member.
(61) In some implementations, the corner guards **160** are completely within the datum structure **130**. In other words, the corner guards **160** are interior rather than exterior to the frame **120**, e.g., enclosed by portions **166** of the frame **120**. The angled sheet of lead **168** can be chamfered at an acute angle θ, e.g., 30° or 60°, relative to the portions **166**. The corner guards can be placed in each of the four corners within the X-ray shielding device **100**.
(62) Referring to FIGS. **3**B and **7**, the gap between shielding panels **132***a* and **132***b* can be reduced by using slots as the shape for holes **134**. Using slots allows for biasing the shielding panels **132** up against a blast wall, indicated by the dashed line in FIG. **7**. For example, shielding panel **132***a* can slide further to the left, e.g., along the X axis, to reduce the gap between the blast wall, e.g., the interior side of the X-ray shielding device **100** receiving X-ray radiation directly from the X-ray source **101**. Additionally, the metal piece, e.g., C-channel **150**, can compensate, e.g., provide additional shielding, for any extra spacing created on the right side of the shielding panel **132***a* where it meets another shielding panel on the same face. Accordingly, in tandem, the datum structure **130** and the metal piece provide flexible alignment without reducing the shielding efficiency of the X-ray shielding device **100**.
(63) Although implementations with a C-channel for the metal piece and doubly chamfered corner guard have been described, other implementations are possible. For example, the metal piece can have a square, rectangular, or circular shape. As another example, the corner guard can have a rectangular shape with one corner being chamfered.
(64) Embodiments of the subject matter and the functional operations described in this specification can be implemented in digital electronic circuitry, or in computer software, firmware, or hardware, including the structures disclosed in this specification and their structural equivalents, or in combinations of one or more of them. Embodiments of the subject matter described in this specification can be implemented using one or more modules of computer program instructions encoded on a non-transitory computer-readable medium for execution by, or to control the operation of, data processing apparatus. The computer-readable medium can be a manufactured product, such as a hard drive in a computer system or an optical disc sold through retail channels, or an embedded system. The computer-readable medium can be acquired separately and later encoded with the one or more modules of computer program instructions, such as by delivery of the one or more modules of computer program instructions over a wired or wireless network. The computer-readable medium can be a machine-readable storage device, a machine-readable storage substrate, a memory device, or a combination of one or more of them.

(65) The term “data processing apparatus” encompasses all apparatus, devices, and machines for processing data, including by way of example a programmable processor, a computer, or multiple processors or computers. The apparatus can include, in addition to hardware, code that creates an execution environment for the computer program in question, e.g., code that constitutes processor firmware, a protocol stack, a database management system, an operating system, a runtime environment, or a combination of one or more of them. In addition, the apparatus can employ various different computing model infrastructures, such as web services, distributed computing and grid computing infrastructures.
(66) A computer program (also known as a program, software, software application, script, or code) can be written in any form of programming language, including compiled or interpreted languages, declarative or procedural languages, and it can be deployed in any form, including as a stand-alone program or as a module, component, subroutine, or other unit suitable for use in a computing environment. A computer program does not necessarily correspond to a file in a file system. A program can be stored in a portion of a file that holds other programs or data (e.g., one or more scripts stored in a markup language document), in a single file dedicated to the program in question, or in multiple coordinated files (e.g., files that store one or more modules, sub-programs, or portions of code). A computer program can be deployed to be executed on one computer or on multiple computers that are located at one site or distributed across multiple sites and interconnected by a communication network.
(67) The processes and logic flows described in this specification can be performed by one or more programmable processors executing one or more computer programs to perform functions by operating on input data and generating output. The processes and logic flows can also be performed by, and apparatus can also be implemented as, special purpose logic circuitry, e.g., an FPGA (field programmable gate array) or an ASIC (application-specific integrated circuit).
(68) Processors suitable for the execution of a computer program include, by way of example, both general and special purpose microprocessors, and any one or more processors of any kind of digital computer. Generally, a processor will receive instructions and data from a read-only memory or a random access memory or both. The essential elements of a computer are a processor for performing instructions and one or more memory devices for storing instructions and data. Generally, a computer will also include, or be operatively coupled to receive data from or transfer data to, or both, one or more mass storage devices for storing data, e.g., magnetic, magneto-optical disks, or optical disks. However, a computer need not have such devices. Moreover, a computer can be embedded in another device, e.g., a mobile telephone, a personal digital assistant (PDA), a mobile audio or video player, a game console, a Global Positioning System (GPS) receiver, or a portable storage device (e.g., a universal serial bus (USB) flash drive), to name just a few. Devices suitable for storing computer program instructions and data include all forms of non-volatile memory, media and memory devices, including by way of example semiconductor memory devices, e.g., EPROM (Erasable Programmable Read-Only Memory), EEPROM (Electrically Erasable Programmable Read-Only Memory), and flash memory devices; magnetic disks, e.g., internal hard disks or removable disks; magneto-optical disks; and CD-ROM and DVD-ROM disks. The processor and the memory can be supplemented by, or incorporated in, special purpose logic circuitry.
(69) To provide for interaction with a user, embodiments of the subject matter described in this specification can be implemented on a computer having a display device, e.g., an LCD (liquid crystal display) display device, an OLED (organic light emitting diode) display device, or another monitor, for displaying information to the user, and a keyboard and a pointing device, e.g., a mouse or a trackball, by which the user can provide input to the computer. Other kinds of devices can be used to provide for interaction with a user as well; for example, feedback provided to the user can be any form of sensory feedback, e.g., visual feedback, auditory feedback, or tactile feedback; and input from the user can be received in any form, including acoustic, speech, or tactile input.

(70) The computing system can include clients and servers. A client and server are generally remote from each other and typically interact through a communication network. The relationship of client and server arises by virtue of computer programs running on the respective computers and having a client-server relationship to each other. Embodiments of the subject matter described in this specification can be implemented in a computing system that includes a back-end component, e.g., as a data server, or that includes a middleware component, e.g., an application server, or that includes a front-end component, e.g., a client computer having a graphical user interface or a Web browser through which a user can interact with an implementation of the subject matter described is this specification, or any combination of one or more such back-end, middleware, or front-end components. The components of the system can be interconnected by any form or medium of digital data communication, e.g., a communication network. Examples of communication networks include a local area network ("LAN") and a wide area network ("WAN"), an inter-network (e.g., the Internet), and peer-to-peer networks (e.g., ad hoc peer-to-peer networks).
(71) While this specification contains many implementation details, these should not be construed as limitations on the scope of what is being or may be claimed, but rather as descriptions of features specific to particular embodiments of the disclosed subject matter. Certain features that are described in this specification in the context of separate embodiments can also be implemented in combination in a single embodiment. Conversely, various features that are described in the context of a single embodiment can also be implemented in multiple embodiments separately or in any suitable subcombination. Moreover, although features may be described above as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination can in some cases be excised from the combination, and the claimed combination may be directed to a subcombination or variation of a subcombination.
(72) Similarly, while operations are depicted in the drawings in a particular order, this should not be understood as requiring that such operations be performed in the particular order shown or in sequential order, or that all illustrated operations be performed, to achieve desired results. In certain circumstances, multitasking and parallel processing may be advantageous. Moreover, the separation of various system components in the embodiments described above should not be understood as requiring such separation in all embodiments, and it should be understood that the described program components and systems can generally be integrated together in a single software product or packaged into multiple software products.
(73) Thus, particular embodiments of the invention have been described. Other embodiments are within the scope of the following claims.

Claims

1. An X-ray device comprising: an X-ray source configured to emit X-rays; a scintillator arranged to absorb the X-rays after interaction with an object that has been placed in the X-ray device, the scintillator being configured to emit light in response to absorption of the X-rays; a detector arranged to receive the light from the scintillator; a frame enclosing the X-ray source and the detector; standoffs positioned on the frame, wherein each of the standoffs has a length and a cross-sectional size; shielding panels comprising lead, wherein each of the shielding panels includes holes having a cross-sectional size that is greater than the cross-sectional size of the standoffs by an amount that ensures the standoffs will pass through the holes despite variations in a placement of the holes in a given shielding panel resulting from manufacturing of the given shielding panel, and the length of the standoffs is greater than a maximum thickness possible for the given shielding panel due to variations in thickness resulting from the manufacturing of the given shielding panel; one or more brackets with fasteners configured to attach to the standoffs, wherein the standoffs form a datum structure for the one or more brackets; and exterior panels with fasteners configured to attach to the one or more brackets.

2. The X-ray device of claim 1, comprising at least one additional shielding panel without lead.
3. The X-ray device of claim 1, further comprising at least one corner guard comprising an angled sheet of lead extending between two adjacent sides of the frame, wherein the angled sheet of lead is placed at an acute angle between the two adjacent sides of the shielding panels, and edges of the angled sheet of lead is chamfered in accordance with the acute angle.
4. The X-ray device of claim 3, further comprising a metal sheet on which the corner guard is attached, wherein the metal sheet is attached to the frame.
5. The X-ray device of claim 1, wherein each of the shielding panels is a laminate comprising lead and steel.
6. The X-ray device of claim 1, wherein at least one of the one or more brackets is configured and arranged to have two of the exterior panels located on different sides of the X-ray device attached to a same bracket.
7. The X-ray device of claim 1, comprising one or more additional shielding panels comprising a plastic impregnated with lead-free particles.
8. The X-ray device of claim 1, wherein each of the standoffs comprises a threaded hole, and each of the fasteners is configured to attach the brackets to the standoffs is a bolt configured to mate with the threaded hole.
9. The X-ray device of claim 1, wherein each shielding panel of the shielding panels weighs less than 39 kilograms.
10. The X-ray device of claim 1, wherein the cross-sectional size of the holes is greater than the cross-sectional size of the standoffs by about 35%.
11. The X-ray device of claim 1, wherein a shape of the holes in the shielding panels is a rounded rectangle.
12. The X-ray device of claim 1, wherein the standoffs are arranged on at least two sides of the frame, thereby forming the datum structure in at least two intersecting planes of three-dimensional space.
13. The X-ray device of claim 1, wherein neighboring edges of first and second shielding panels of the shielding panels form a gap, and further comprising: a metal piece having fasteners configured to attach the metal piece with the frame, wherein the metal piece is shaped to receive a shielding piece comprising lead; and the shielding piece being sized and positioned to prevent X-rays from passing (i) through the gap and (ii) through holes in the first and second shielding panels on either side of the gap, which are usable to couple the first and second shielding panels with the frame.
14. An X-ray device comprising: an X-ray source configured to emit X-rays; a scintillator arranged to absorb the X-rays after interaction with an object that has been placed in the X-ray device, the scintillator being configured to emit light in response to absorption of the X-rays; a detector arranged to receive the light from the scintillator; a frame enclosing the X-ray source and the detector; shielding panels comprising lead, wherein each of the shielding panels includes holes usable when coupling the shielding panel with the frame, wherein a first and a second of the shielding panels protect a single side of the frame and have been reduced in size to facilitate installation of the first and second shielding panels, and wherein a gap remains between the first shielding panel and the second shielding panel when installed on the single side of the frame; a metal piece having fasteners configured to attach the metal piece with the frame, wherein the metal piece is shaped to receive a shielding piece comprising lead; and the shielding piece being sized and positioned to prevent X-rays from passing (i) through the gap and (ii) through the holes in the first and second shielding panels on either side of the gap, which are usable to couple the first and second shielding panels with the frame.
15. The X-ray device of claim 14, further comprising additional shielding disposed on the shielding.
16. The X-ray device of claim 14, further comprising corner guards comprising angled sheets of lead extending between two adjacent faces of the shielding panels, wherein the corner guards are

angled at an acute angle between the two adjacent faces of the shielding panels, and edges of the corner guards are chamfered at the acute angle.

17. The X-ray device of claim 16, further comprising a sheet on which each corner guard of the corner guards is attached, wherein the sheet is attached to the frame.

18. The X-ray device of claim 14, wherein the shielding panels comprise at least one laminate comprising steel and lead.

19. The X-ray device of claim 14, further comprising brackets and standoffs, wherein the standoffs comprise threaded holes, and fasteners that attach the brackets to the standoffs are bolts configured to mate with the threaded holes in the standoffs.

20. The X-ray device of claim 19, wherein a cross-sectional size of the threaded holes is greater than a cross-sectional size of the standoffs by about 35%.
